

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021- 2022 (Even Semester)
Active Learning Practices/ Content beyond Syllabus

Degree, Semester& Branch: VIII Semester B.E.ECE A

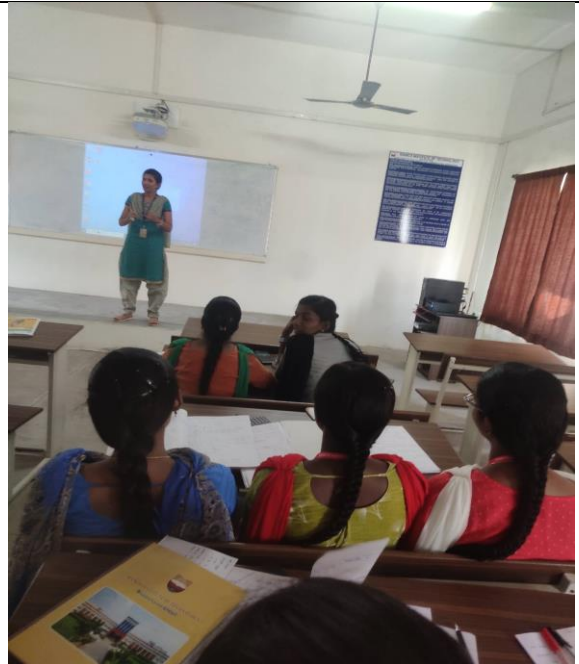
Course Code & Title: EC8094 Satellite Communication

Name of the Faculty member: Mr.R.Deiva Nayagam

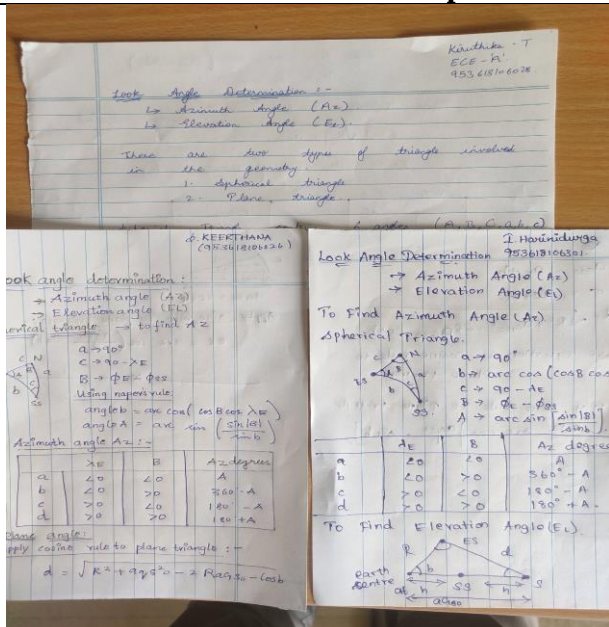
Active Learning Practices Execution

UNIT I SATELLITE ORBITS

Activity: Zero Minute Speech
Topic: Frequency Allocation of Satellite Services



Activity: One Minute Paper
Topic: Look Angle Determination



Activity: Mindmap
Topic: Launching Procedures



Content Beyond Syllabus
Topic: Launch Vehicle Design and Analysis



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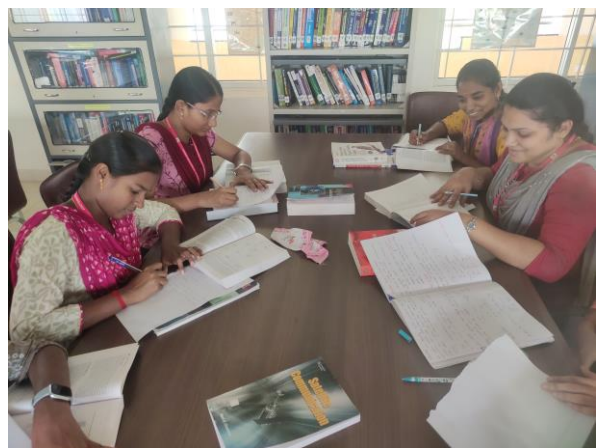
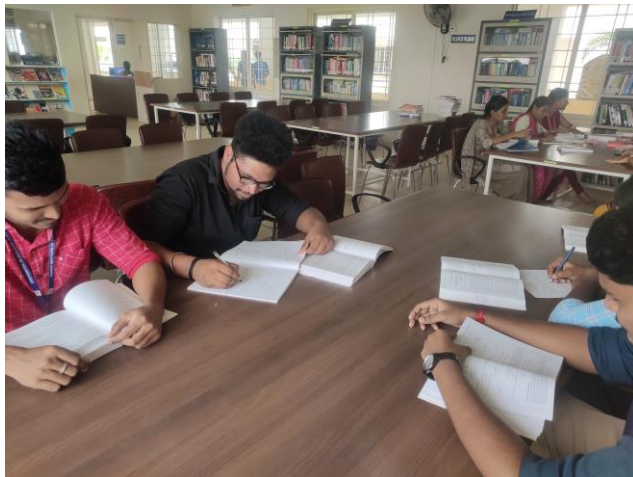
Name of the Faculty member: Mr.R.Deiva Nayagam

Active Learning Practices Execution

UNIT II SPACE SEGMENT

Activity: Open Book Test (Library Visit)

Topic: Telemetry, Tracking and Command Subsystem



Activity: Laboratory Visit

Topic: Antenna Subsystem





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Course Code & Title: EC8094 Satellite Communication

Name of the Faculty member: Mr.R.Deiva Nayagam, AP/ECE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Code Division Multiple Access
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO12
- **Activity Chosen:** JigSaw
- **Justification:**

The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal; each person depends on all the others.

The basic concept of multiple access scheme was already learnt in Wireless Communication. The topic is related to the same concepts and easy to understand.

- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

- **Materials for the Activity:**

The materials for the preparation of the students will be shared one week before through LMS Canvas. Including the material, students were asked to refer to the following book and web content also.

Book Title/ Author/ Publisher/ Edition	Page No.
1. Dennis Roddy, "Satellite Communication", 4th Edition, Tata McGraw Hill International, 2006.	472-483
Websites	
1. https://ocw.mit.edu/courses/6-450-principles-of-digital-communications-i-fall-2006/resources/lecture-24-case-study-on-code-division-multiple-access/	
2. https://www.jigsaw.org/	
3. https://www.newcastle.edu.au/data/assets/pdf_file/0016/109600/Jigsaw-learningactivity.pdf	

- The topic on CDMA was subdivided into five sub categories. The Students are divided into groups of four students.

- Each group is allotted with a name and its known as Expert group EC1, EC2, EC3, EC4 & EC5
- Each expert group is allotted with one topic
 - **Direct sequence spread spectrum**
 - **The code signal $c(t)$**
 - **Acquisition and tracking**
 - **Spectrum spreading and despreading**
 - **CDMA throughput**
- The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas. The Expert group members should go to other group and discuss the points (30 Minutes) to the other members.
- All the members in home group learnt about all the topics through expert group members. Finally one from each group should summarize the points of the topic Fixed and Floating point representation to the class (20 Minutes).
- Review of points and conclusion of the activity (5 minutes)
- Thus the concept of CDMA is known to all by involving the students actively

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO10	PO11	PO12	PSO1	PSO3	PSO4
CO4	2	3	3	2	2	2	1	2	3	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 3	PO 10	PSO 3
	3	2	3
Justification for correlation	Students will function effectively to design the solution for CDMA problems to understand the spreading and despreading problem. So Course outcome is mapped at level 3	In Jigsaw activity students will make effective presentations on Direct Sequence Spread Spectrum concepts, hence Course outcome is mapped at level 2	Spectrum Spreading and Despreading topic used to solve the complex task occurs in Satellite communication problems, So Course outcome is mapped at level 3

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***

Students feel that it was interesting. They learn how to communicate with team members and work together. Students feedback was collected about the concept learned and activity used to learn. All the students replied positively about the activity and the learned concepts.

- ***Benefit of the practice:***

Every students got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test III. Outcome attainment would have increased due to innovative practice over conventional practice. Students easily understand the concepts of Direct Sequence Spread Spectrum. Students can answer easily about Spectrum Spreading and Despreading with and they can design the CDMA throughput using number of channels.

- ***Challenges faced in implementation:***

The main challenge faced is group conflict. Few Students are not ready to work together. Initially, the students not came up to involve in this activity voluntarily. It took a longer time than other strategies.

Faced issues while make the students to understand about Jigsaw rules.

References:

- <https://ocw.mit.edu/courses/6-450-principles-of-digital-communications-i-fall-2006/resources/lecture-24-case-study-on-code-division-multiple-access/>
- Dennis Roddy, "Satellite Communication", 4th Edition, Tata McGraw Hill International, 2006.
- <https://www.youtube.com/watch?v=TJNKPn-G8>



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Course Code & Title: EC8094 Satellite Communication

Name of the Faculty member: Mr.R.Deiva Nayagam, AP/ECE

Innovative Practice Description

- **Unit / Topic:** Unit-II / Antenna Subsystem (Experimental Setup of Horn Antenna)
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO7
- **Activity Chosen:** Laboratory Visit
- **Justification:**

Horn Antenna is used to correlate theoretical concepts of antenna subsystem by doing experimental setup as it is being utilised in Advanced Communication Laboratory. As a result, they must be aware of it. Because basics of Antenna had already studied in the Subject Antennas and Microwave Engineering. A demonstration is a step-by-step technique for showing someone how to measure the range of frequency.

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

(i) Materials for the Activity:

The materials for the preparation of the students will be shared one week before through LMS Canvas. Including the material, students were asked to refer the book and web content also.

References:

Book Title/ Author/ Publisher/ Edition	Page No.
1. Dennis Roddy, "Satellite Communication", 4th Edition, Tata McGraw Hill International, 2006.	155-158
2. Timothy, Pratt, Charles, W.Bostain, Jeremy E.Allnutt, "Satellite Communication", 2 nd Edition, Wiley Publications, 2002.	80-83
Websites	
https://www.tutorialspoint.com/antenna_theory/antenna_theory_horn.html	
https://electronicsdesk.com/horn-antenna.html	
https://www.antenna-theory.com/antennas/aperture/horn.php	

- Laboratory Visit is a specific type of fulfilling the gap in theoretical class that work best when used with the whole class. The class is first divided into groups of four to six learners who are then given some information on a particular aspect of the topic.
- The class students is then organised in Laboratory of five students in agroup so that each group members can do the experiments in real time setup of antenna subsystem using Horn Antenna.
- Before entering into the laboratory, the topic on antenna system and design aspect of horn antenna was taught.
- Then the students were moved to visit the laboratory to see the experimental setup of horn antenna and take the readings in various angles to measure the variation in amplitude and frequency between transmitter and receiver antenna.
- Thus the concept of Horn antenna setup was known to all by involving the students actively

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO10	PO11	PO12	PSO1	PSO3
CO4	2	3	1	1	2	2	1	1	2	3
(1 – Low			2 – Moderate			3 – High)				

• **PO / PSO mapped**

Innovative practice	PO 5	PO 12	PSO 3
	2	1	3
Justification for correlation	Students used the hardware demonstration of horn antenna setup to understand the demonstration about how to measure the range of frequency and analyze its variations between transmitter and receiver antenna	Students can understand the knowledge of antenna subsystems and they can utilize the knowledge in designing of antenna to receive the various signals depends on the applications	The students can able to design, analyze and develop the antenna design model for the field of communication systems

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***

The feedback was collected from the students about the activity for learning the concept. All of them really interested to attend the session on engaging them with real time experimental setup of horn antenna. The students really enjoyed to learn the concepts on antenna subsystem and understand the impact on the signal received between transmitting and receiving horn antenna

- ***Benefit of the practice:***

Students easily understand the concepts of antenna subsystem with its experimental setup. The success of the activity was evaluated by asking the same question in Internal Assessment test II. Outcome attainment would have increased due to innovative practice over conventional practice.

- ***Challenges faced in implementation:***

The activity should not be able to complete in the planned duration.

References:

- Dennis Roddy, "Satellite Communication", 4th Edition, Tata McGraw Hill International, 2006.
- Timothy, Pratt, Charles, W.Bostain, Jeremy E.Allnutt, "Satellite Communication", 2nd Edition, Wiley Publications, 2002.
- <https://www.antenna-theory.com/antennas/aperture/horn.php>